Machine Learning

Dataset Description – Student Performance ML Dataset

The dataset `student_performance_ml.csv` contains academic and behavioral information of students. The objective of this dataset is to predict whether a student will **Pass (1)** or **Fail (0)** based on various input features.

Each row in the dataset represents one student, and each column represents a measurable factor that may influence academic performance.

Features Description

- **StudyHours** – Number of hours a student studies per day.
- **Attendance** – Percentage of class attendance.
- **PreviousScore** – Marks obtained in the previous examination.
- **AssignmentsCompleted** – Number of assignments completed by the student.
- **SleepHours** – Average number of hours the student sleeps per day.
- **FinalResult** – Target variable (Output):
  - `1` → Pass
  - `0` → Fail

Objective of the Dataset

The goal is to:

- Analyze how different factors affect student performance.
- Build a Machine Learning model to predict whether a student will pass or fail.
- Understand concepts such as training, testing, accuracy, confusion matrix, overfitting, and model evaluation.

1. Write a Python program to load the file `student_performance_ml.csv` using pandas. Display:

- First 5 records
- Last 5 records
- Total number of rows and columns
- List of column names
- Data types of each column

2. Write a program to:

- Display total number of students in the dataset
- Count how many students Passed (FinalResult = 1)
- Count how many students Failed (FinalResult = 0)

3. Using pandas functions, calculate and display:

- Average StudyHours
- Average Attendance
- Maximum PreviousScore
- Minimum SleepHours

4. Use `value_counts()` to analyze the distribution of FinalResult.
Calculate the percentage of Pass and Fail students.
Is the dataset balanced? Justify your answer.

5. Based on the dataset values, analyze whether:

- Higher StudyHours increase the chance of passing.
- Higher Attendance improves FinalResult.
  Write your observations in 4–5 lines.

6. Plot a histogram of StudyHours.
Explain what the distribution tells you.

7. Create a scatter plot of:
StudyHours vs PreviousScore

Use different colors for Pass and Fail students.

8. Draw a boxplot for Attendance.
Identify if any outliers are present.

9. Create a plot showing relationship between AssignmentsCompleted and FinalResult.
Explain your observation.

10. Plot SleepHours against FinalResult.
Does sleeping more guarantee success? Explain.